06_lag_diagnostics_descriptive

May 11, 2026

1 06 Lag Diagnostics (Descriptive)

This notebook runs modest lag diagnostics on the combined monthly salience-polling panel.

Scope: - Descriptive only. - No causal claims. - No significance testing or event-study regression.

[1]:
```
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option('display.max_columns', 200)
pd.set_option('display.width', 180)

PROJECT_ROOT = Path.cwd()
if PROJECT_ROOT.name == 'notebooks':
    PROJECT_ROOT = PROJECT_ROOT.parent

PANEL_DIR = PROJECT_ROOT / 'data' / 'panel'
FIGURES_DIR = PROJECT_ROOT / 'figures' / '06_notebook_figures'
FIGURES_DIR.mkdir(parents=True, exist_ok=True)

PANEL_PARQUET = PANEL_DIR / 'monthly_salience_polling_panel_combined_2013_2025.
↪parquet'
PANEL_CSV = PANEL_DIR / 'monthly_salience_polling_panel_combined_2013_2025.csv'
OUT_LAG_CSV = PANEL_DIR / 'lag_diagnostics_descriptive_summary.csv'

EXPECTED_START = pd.Timestamp('2013-09-01')
EXPECTED_END = pd.Timestamp('2025-12-31')
EXPECTED_ROWS = 148

print('PANEL_PARQUET:', PANEL_PARQUET)
print('OUT_LAG_CSV:', OUT_LAG_CSV)
print('FIGURES_DIR:', FIGURES_DIR)
```

```
PANEL_PARQUET: c:\PythonProjects\AfD-
project\data\panel\monthly_salience_polling_panel_combined_2013_2025.parquet
OUT_LAG_CSV: c:\PythonProjects\AfD-
```

project\data\panel\lag_diagnostics_descriptive_summary.csv
FIGURES_DIR: c:\PythonProjects\AfD-project\figures\06_notebook_figures

[2]:
```
def load_panel(parquet_path: Path, csv_path: Path) -> pd.DataFrame:
    if parquet_path.exists():
        return pd.read_parquet(parquet_path)
    if csv_path.exists():
        return pd.read_csv(csv_path)
    raise FileNotFoundError(f'Missing panel file: {parquet_path} and␣
↪{csv_path}')

panel = load_panel(PANEL_PARQUET, PANEL_CSV).copy()
panel['month_dt'] = pd.to_datetime(panel['month_dt'], errors='coerce')
panel['year_month'] = panel['month_dt'].dt.strftime('%Y-%m')

required_cols = [
    'year_month', 'month_dt', 'afd_poll_support_mean', 'nexis_total_results',
    'polling_source_block', 'n_polls'
]
missing_req = [c for c in required_cols if c not in panel.columns]
if missing_req:
    raise ValueError(f'Missing required columns: {missing_req}')

if 'media_salience_volume_log1p' not in panel.columns and␣
↪'media_salience_log1p' not in panel.columns:
    print('media_salience_volume_log1p not present; will compute from␣
↪nexis_total_results.')

if ('media_salience_volume_z_panel' not in panel.columns) and␣
↪('media_salience_volume_z' not in panel.columns):
    raise ValueError('Missing both media_salience_volume_z_panel and␣
↪media_salience_volume_z')

if panel['afd_poll_support_mean'].isna().any():
    raise ValueError('afd_poll_support_mean contains missing values')
if panel['nexis_total_results'].isna().any():
    raise ValueError('nexis_total_results contains missing values')
if panel['month_dt'].isna().any():
    raise ValueError('month_dt contains parse failures')
if panel['year_month'].duplicated().any():
    raise ValueError('Panel has duplicate months')

panel = panel.sort_values('month_dt').reset_index(drop=True)

if panel['month_dt'].min().to_period('M') != EXPECTED_START.to_period('M') or␣
↪panel['month_dt'].max().to_period('M') != EXPECTED_END.to_period('M'):
```

```
    raise ValueError(f'Unexpected coverage: {panel["month_dt"].min()} to␣
↪{panel["month_dt"].max()}')
if len(panel) != EXPECTED_ROWS:
    raise ValueError(f'Unexpected row count: {len(panel)} (expected␣
↪{EXPECTED_ROWS})')

if 'retrieval_mismatch_flag' in panel.columns:
    mismatch_count = int(panel['retrieval_mismatch_flag'].fillna(False).
↪astype(bool).sum())
    if mismatch_count != 0:
        raise ValueError(f'retrieval_mismatch_flag has {mismatch_count} true␣
↪values')

print('Validation passed.')
print('shape:', panel.shape)
print('coverage:', panel['month_dt'].min(), 'to', panel['month_dt'].max())
print('source blocks:', panel['polling_source_block'].value_counts().to_dict())
```

```
Validation passed.
shape: (148, 41)
coverage: 2013-09-01 00:00:00 to 2025-12-01 00:00:00
source blocks: {'dawum': 108, 'historical_manual_wahlrecht': 40}
```

[3]:
```
def zscore(series: pd.Series) -> pd.Series:
    s = pd.to_numeric(series, errors='coerce')
    mu = s.mean()
    sd = s.std(ddof=0)
    if pd.isna(sd) or sd == 0:
        return pd.Series(np.zeros(len(s)), index=s.index, dtype=float)
    return (s - mu) / sd

panel['afd_poll_support_change'] = panel['afd_poll_support_mean'].diff()
panel['nexis_total_results_change'] = panel['nexis_total_results'].diff()

if 'media_salience_volume_log1p' in panel.columns:
    panel['media_salience_log1p'] = pd.
↪to_numeric(panel['media_salience_volume_log1p'], errors='coerce')
elif 'media_salience_log1p' in panel.columns:
    panel['media_salience_log1p'] = pd.
↪to_numeric(panel['media_salience_log1p'], errors='coerce')
else:
    panel['media_salience_log1p'] = np.log1p(pd.
↪to_numeric(panel['nexis_total_results'], errors='coerce'))

panel['media_salience_log1p_change'] = panel['media_salience_log1p'].diff()
panel['afd_poll_support_z_combined'] = zscore(panel['afd_poll_support_mean'])
```

```python
if 'media_salience_volume_z_panel' in panel.columns:
    panel['media_salience_z_combined'] = pd.
 ↪to_numeric(panel['media_salience_volume_z_panel'], errors='coerce')
else:
    panel['media_salience_z_combined'] = pd.
 ↪to_numeric(panel['media_salience_volume_z'], errors='coerce')

for lag in range(0, 7):
    panel[f'media_salience_z_lag{lag}'] = panel['media_salience_z_combined'].
 ↪shift(lag)
    panel[f'media_salience_log1p_lag{lag}'] = panel['media_salience_log1p'].
 ↪shift(lag)
    panel[f'nexis_total_results_lag{lag}'] = panel['nexis_total_results'].
 ↪shift(lag)

print(panel[['year_month','afd_poll_support_mean','afd_poll_support_change','media_salience_z_
 ↪head(10))
```

```
  year_month  afd_poll_support_mean  afd_poll_support_change
media_salience_z_combined  media_salience_log1p
0    2013-09                  4.775                      NaN
-0.589327              4.564348
1    2013-10                  5.125                    0.350
-0.224905              5.023881
2    2013-11                  4.600                   -0.525
-0.257443              4.990433
3    2013-12                  4.000                   -0.600
-0.355056              4.882802
4    2014-01                  4.000                    0.000
-0.133800              5.111988
5    2014-02                  4.750                    0.750
-0.517744              4.672829
6    2014-03                  4.600                   -0.150
-0.654402              4.454347
7    2014-04                  5.000                    0.400
-0.543774              4.634729
8    2014-05                  5.000                    0.000
-0.276965              4.969813
9    2014-06                  6.000                    1.000
-0.433146              4.787492
```

[4]:
```python
def corr_with_n(df: pd.DataFrame, x: str, y: str) -> tuple[float, int]:
    tmp = df[[x, y]].dropna()
    n = len(tmp)
    if n < 2:
        return (np.nan, n)
    return (tmp[x].corr(tmp[y]), n)
```

```python
samples = {
    'combined_all': panel,
    'historical_manual_wahlrecht': panel[panel['polling_source_block'] ==␣
 ↪'historical_manual_wahlrecht'].copy(),
    'dawum': panel[panel['polling_source_block'] == 'dawum'].copy(),
}

pairs = [
    ('media_salience_z', 'afd_level', 'media_salience_z_lag{lag}',␣
 ↪'afd_poll_support_mean'),
    ('media_salience_z', 'afd_change', 'media_salience_z_lag{lag}',␣
 ↪'afd_poll_support_change'),
    ('media_salience_log1p', 'afd_level', 'media_salience_log1p_lag{lag}',␣
 ↪'afd_poll_support_mean'),
    ('media_salience_log1p', 'afd_change', 'media_salience_log1p_lag{lag}',␣
 ↪'afd_poll_support_change'),
]

rows = []
for sample_name, sdf in samples.items():
    for lag in range(0, 7):
        for x_family, y_family, x_tpl, y_var in pairs:
            x_var = x_tpl.format(lag=lag)
            corr, n_used = corr_with_n(sdf, x_var, y_var)
            rows.append({
                'sample': sample_name,
                'lag_months': lag,
                'x_variable': x_var,
                'y_variable': y_var,
                'correlation': corr,
                'n_used': n_used,
                'notes': 'Descriptive lag correlation only; non-causal and␣
 ↪source-boundary sensitive.'
            })

lag_summary = pd.DataFrame(rows)
lag_summary.to_csv(OUT_LAG_CSV, index=False)
print('Saved lag summary:', OUT_LAG_CSV)
print('lag_summary shape:', lag_summary.shape)
print(lag_summary.head(20))
```

```
Saved lag summary: c:\PythonProjects\AfD-
project\data\panel\lag_diagnostics_descriptive_summary.csv
lag_summary shape: (84, 7)
          sample  lag_months                     x_variable                 y_variable
correlation  n_used                                                              notes
```

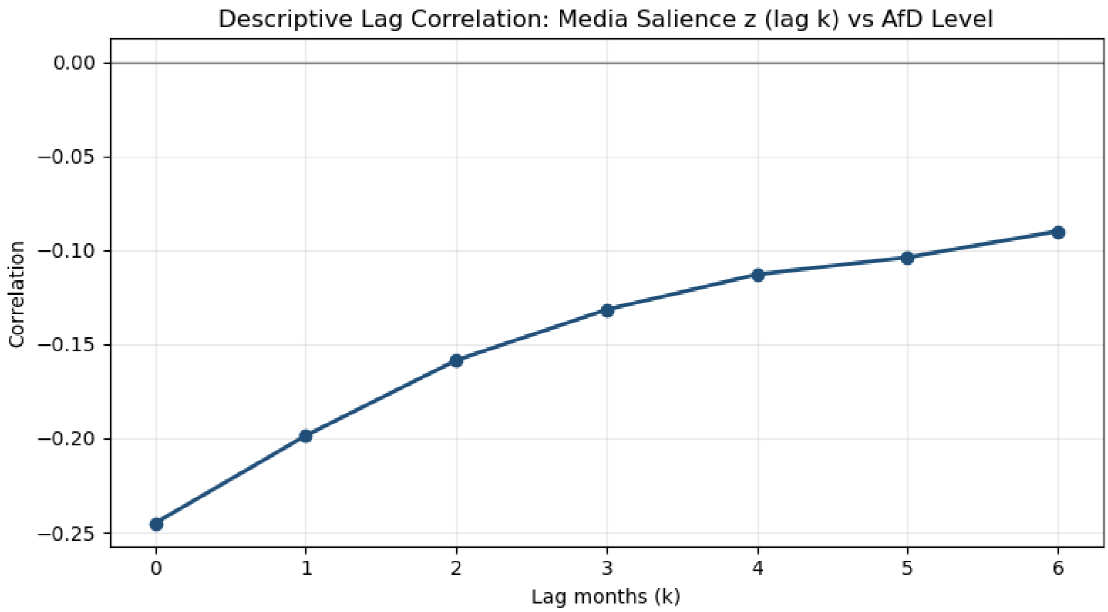
```
0   combined_all           0      media_salience_z_lag0    afd_poll_support_mean
-0.245014     148  Descriptive lag correlation only; non-causal a…
1   combined_all           0      media_salience_z_lag0  afd_poll_support_change
0.165381     147  Descriptive lag correlation only; non-causal a…
2   combined_all           0  media_salience_log1p_lag0    afd_poll_support_mean
-0.176494     148  Descriptive lag correlation only; non-causal a…
3   combined_all           0  media_salience_log1p_lag0  afd_poll_support_change
0.110780     147  Descriptive lag correlation only; non-causal a…
4   combined_all           1      media_salience_z_lag1    afd_poll_support_mean
-0.198573     147  Descriptive lag correlation only; non-causal a…
5   combined_all           1      media_salience_z_lag1  afd_poll_support_change
0.244425     147  Descriptive lag correlation only; non-causal a…
6   combined_all           1  media_salience_log1p_lag1    afd_poll_support_mean
-0.144349     147  Descriptive lag correlation only; non-causal a…
7   combined_all           1  media_salience_log1p_lag1  afd_poll_support_change
0.158191     147  Descriptive lag correlation only; non-causal a…
8   combined_all           2      media_salience_z_lag2    afd_poll_support_mean
-0.158535     146  Descriptive lag correlation only; non-causal a…
9   combined_all           2      media_salience_z_lag2  afd_poll_support_change
0.200125     146  Descriptive lag correlation only; non-causal a…
10  combined_all           2  media_salience_log1p_lag2    afd_poll_support_mean
-0.121611     146  Descriptive lag correlation only; non-causal a…
11  combined_all           2  media_salience_log1p_lag2  afd_poll_support_change
0.095968     146  Descriptive lag correlation only; non-causal a…
12  combined_all           3      media_salience_z_lag3    afd_poll_support_mean
-0.131640     145  Descriptive lag correlation only; non-causal a…
13  combined_all           3      media_salience_z_lag3  afd_poll_support_change
0.114839     145  Descriptive lag correlation only; non-causal a…
14  combined_all           3  media_salience_log1p_lag3    afd_poll_support_mean
-0.103559     145  Descriptive lag correlation only; non-causal a…
15  combined_all           3  media_salience_log1p_lag3  afd_poll_support_change
0.055323     145  Descriptive lag correlation only; non-causal a…
16  combined_all           4      media_salience_z_lag4    afd_poll_support_mean
-0.112942     144  Descriptive lag correlation only; non-causal a…
17  combined_all           4      media_salience_z_lag4  afd_poll_support_change
0.064918     144  Descriptive lag correlation only; non-causal a…
18  combined_all           4  media_salience_log1p_lag4    afd_poll_support_mean
-0.089097     144  Descriptive lag correlation only; non-causal a…
19  combined_all           4  media_salience_log1p_lag4  afd_poll_support_change
0.033003     144  Descriptive lag correlation only; non-causal a…
```

[5]:

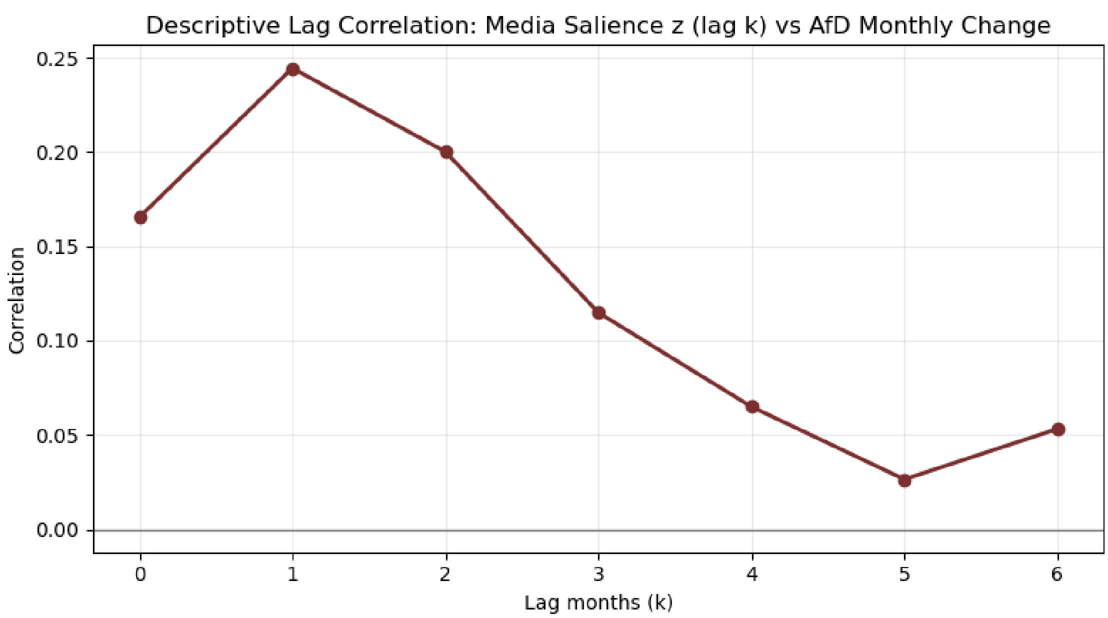
```python
# Figure 1: combined lag correlations (media z vs AfD level)
plot1 = lag_summary[(lag_summary['sample'] == 'combined_all') &␣
 ↪(lag_summary['x_variable'].str.startswith('media_salience_z_lag')) &␣
 ↪(lag_summary['y_variable'] == 'afd_poll_support_mean')].copy()
plot1 = plot1.sort_values('lag_months')
```

```
plt.figure(figsize=(8, 4.5))
plt.plot(plot1['lag_months'], plot1['correlation'], marker='o', linewidth=2,␣
↪color='#1f4e79')
plt.axhline(0, color='gray', linewidth=1)
plt.title('Descriptive Lag Correlation: Media Salience z (lag k) vs AfD Level')
plt.xlabel('Lag months (k)')
plt.ylabel('Correlation')
plt.grid(alpha=0.25)
out1 = FIGURES_DIR / 'lag_correlations_media_z_vs_afd_level.png'
plt.tight_layout(); plt.savefig(out1, dpi=150); plt.show()
print('Saved:', out1)

# Figure 2: combined lag correlations (media z vs AfD change)
plot2 = lag_summary[(lag_summary['sample'] == 'combined_all') &␣
↪(lag_summary['x_variable'].str.startswith('media_salience_z_lag')) &␣
↪(lag_summary['y_variable'] == 'afd_poll_support_change')].copy()
plot2 = plot2.sort_values('lag_months')

plt.figure(figsize=(8, 4.5))
plt.plot(plot2['lag_months'], plot2['correlation'], marker='o', linewidth=2,␣
↪color='#7a2e2e')
plt.axhline(0, color='gray', linewidth=1)
plt.title('Descriptive Lag Correlation: Media Salience z (lag k) vs AfD Monthly␣
↪Change')
plt.xlabel('Lag months (k)')
plt.ylabel('Correlation')
plt.grid(alpha=0.25)
out2 = FIGURES_DIR / 'lag_correlations_media_z_vs_afd_change.png'
plt.tight_layout(); plt.savefig(out2, dpi=150); plt.show()
print('Saved:', out2)

# Figure 3: by source block, media z lag vs AfD level
plot3 = lag_summary[(lag_summary['sample'].
↪isin(['historical_manual_wahlrecht','dawum'])) & (lag_summary['x_variable'].
↪str.startswith('media_salience_z_lag')) & (lag_summary['y_variable'] ==␣
↪'afd_poll_support_mean')].copy()
plot3 = plot3.sort_values(['sample','lag_months'])

plt.figure(figsize=(8.5, 4.8))
for s, color in [('historical_manual_wahlrecht', '#8a1c1c'), ('dawum',␣
↪'#1c4b8a')]:
    sub = plot3[plot3['sample'] == s]
    label = f"{s} (n range {int(sub['n_used'].min())}-{int(sub['n_used'].
↪max())})"
    plt.plot(sub['lag_months'], sub['correlation'], marker='o', linewidth=2,␣
↪label=label, color=color)
```

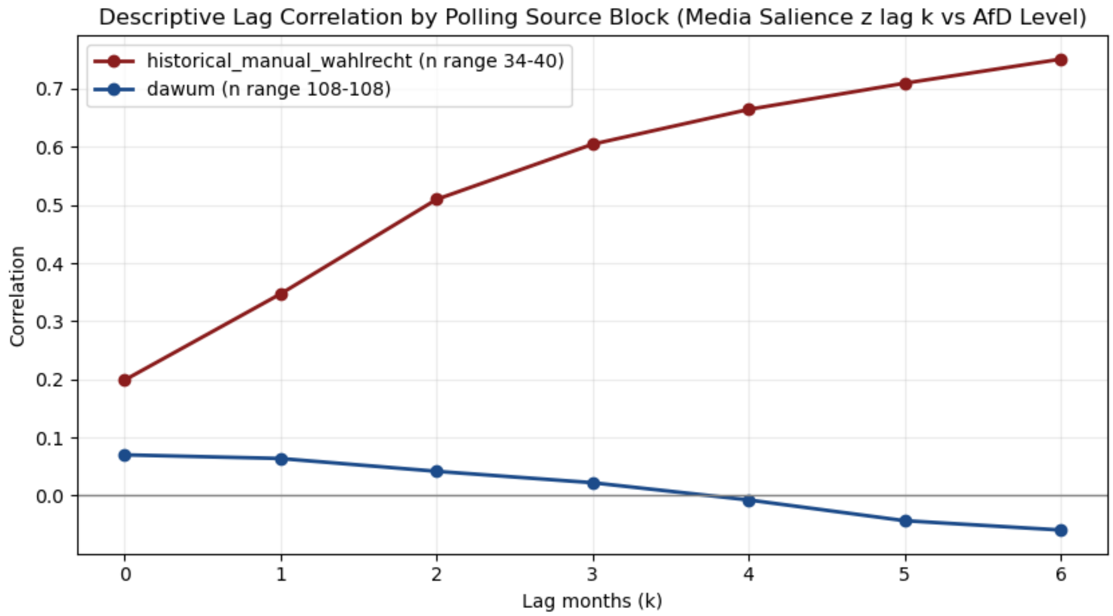
```
plt.axhline(0, color='gray', linewidth=1)
plt.title('Descriptive Lag Correlation by Polling Source Block (Media Salience␣
↪z lag k vs AfD Level)')
plt.xlabel('Lag months (k)')
plt.ylabel('Correlation')
plt.grid(alpha=0.25)
plt.legend(loc='best')
out3 = FIGURES_DIR / 'lag_correlations_by_source_block.png'
plt.tight_layout(); plt.savefig(out3, dpi=150); plt.show()
print('Saved:', out3)

# Figure 4: media salience and AfD change over time (descriptive)
fig, ax1 = plt.subplots(figsize=(12, 4.8))
ax1.plot(panel['month_dt'], panel['media_salience_z_combined'],␣
↪color='#1f4e79', linewidth=1.8, label='Media salience z')
ax1.set_ylabel('Media salience z', color='#1f4e79')
ax1.tick_params(axis='y', labelcolor='#1f4e79')
ax1.grid(alpha=0.2)

ax2 = ax1.twinx()
ax2.plot(panel['month_dt'], panel['afd_poll_support_change'], color='#7a2e2e',␣
↪linewidth=1.3, label='AfD monthly change')
ax2.set_ylabel('AfD monthly change (pp)', color='#7a2e2e')
ax2.tick_params(axis='y', labelcolor='#7a2e2e')

ax1.set_title('Descriptive Series: Media Salience and AfD Monthly Polling␣
↪Change')
ax1.set_xlabel('Month')

h1,l1 = ax1.get_legend_handles_labels(); h2,l2 = ax2.get_legend_handles_labels()
ax1.legend(h1+h2, l1+l2, loc='upper left')
out4 = FIGURES_DIR / 'media_salience_and_afd_change_descriptive.png'
fig.tight_layout(); fig.savefig(out4, dpi=150); plt.show()
print('Saved:', out4)
```



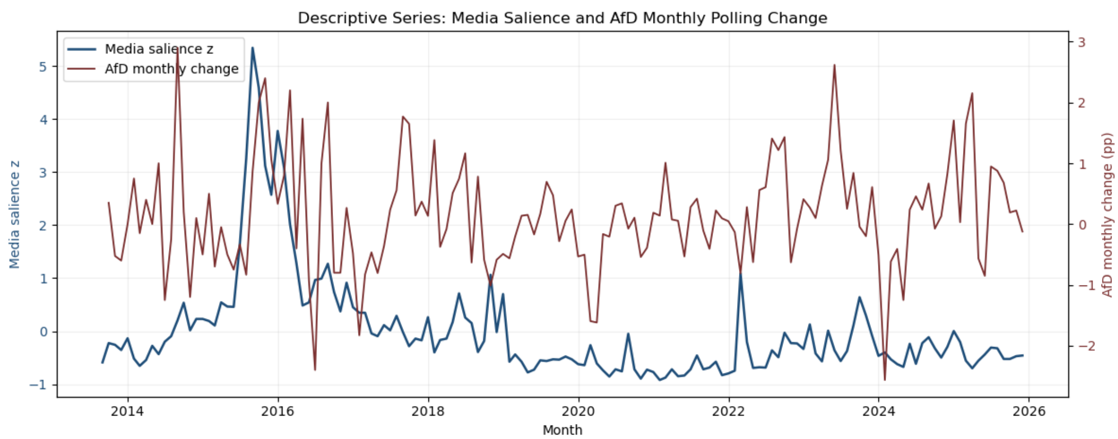

```
Saved: c:\PythonProjects\AfD-
project\figures\06_notebook_figures\lag_correlations_media_z_vs_afd_level.png
```




```
Saved: c:\PythonProjects\AfD-
project\figures\06_notebook_figures\lag_correlations_media_z_vs_afd_change.png
```



Saved: c:\PythonProjects\AfD-project\figures\06_notebook_figures\lag_correlations_by_source_block.png



Saved: c:\PythonProjects\AfD-project\figures\06_notebook_figures\media_salience_and_afd_change_descriptive.png

[6]:

```
# Compact textual summary of key lag patterns
summary_level = lag_summary[(lag_summary['sample'] == 'combined_all') &
 ↪(lag_summary['x_variable'].str.startswith('media_salience_z_lag')) &
 ↪(lag_summary['y_variable'] == 'afd_poll_support_mean')].
 ↪sort_values('correlation', ascending=False)
```

```python
summary_change = lag_summary[(lag_summary['sample'] == 'combined_all') &
    (lag_summary['x_variable'].str.startswith('media_salience_z_lag')) &
    (lag_summary['y_variable'] == 'afd_poll_support_change')].
    sort_values('correlation', ascending=False)

print('Top 3 correlations: media z lag vs AfD level (combined_all)')
print(summary_level[['lag_months','correlation','n_used']].head(3).
    to_string(index=False))
print('Top 3 correlations: media z lag vs AfD change (combined_all)')
print(summary_change[['lag_months','correlation','n_used']].head(3).
    to_string(index=False))

print('By source block, lag0 media z vs AfD level:')
for s in ['historical_manual_wahlrecht','dawum']:
    row = lag_summary[(lag_summary['sample']==s) &
    (lag_summary['lag_months']==0) &
    (lag_summary['x_variable']=='media_salience_z_lag0') &
    (lag_summary['y_variable']=='afd_poll_support_mean')]
    print(s, row[['correlation','n_used']].to_dict(orient='records'))
```

```
Top 3 correlations: media z lag vs AfD level (combined_all)
 lag_months  correlation  n_used
          6    -0.090020     142
          5    -0.103852     143
          4    -0.112942     144
Top 3 correlations: media z lag vs AfD change (combined_all)
 lag_months  correlation  n_used
          1     0.244425     147
          2     0.200125     146
          0     0.165381     147
By source block, lag0 media z vs AfD level:
historical_manual_wahlrecht [{'correlation': 0.19877876320088197, 'n_used': 40}]
dawum [{'correlation': 0.07019560682689899, 'n_used': 108}]
```

1.1 Interpretation (Descriptive)

- These lag diagnostics are exploratory.
- They inspect whether media salience tends to lead, coincide with, or lag AfD polling levels/changes in a simple descriptive sense.
- AfD support has strong long-run trends, and the combined panel has a polling-source boundary at 2017-01.
- Correlations should therefore not be interpreted causally.
- Use these outputs to guide later modeling choices, not as final evidence.